Automated Formal verification of ShEx with Lean

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Abstract. Validation of RDF data is a key task for the Semantic Web and Knowledge Graphs. Shape Expressions (ShEx) is a popular language for RDF validation that has been presented with a formal semantics in several academic papers. We present a Lean 4 representation of ShEx based on two papers that can be used to verify the correctness of the language and the validation algorithms presented in those papers. The development of this encoding can be useful to safely extend the language with new features.

Keywords: RDF · ShEx · Shape Expressions

1 Introduction

RDF is a key technology for the Semantic Web and Knowledge Graphs. Describing and validating RDF data has been an important topic in the last decade with two main technologies proposed: SHACL [10] and ShEx [17]. SHACL (Shapes Constraint Language) is a W3C recommendation whose definition was based on an informal semantics in natural language that left several features like recursion undefined. On the other hand, ShEx followed a community-driven approach [16] whose semantic specification was based on the formal semantics presented in several papers: [18] introduced the complexity of the basic validation algorithm, [4] presented the semantics of the ShEx schema language with a formal semantics that supported recursion and negation based on stratified semantics, and [5] extended the semantics to support inheritance in ShEx schemas.

Lean 4 [14] has been introduced as an automatic theorem prover based on dependent type theory that can be used to verify the correctness of mathematical proofs. In this paper we present a Lean 4 representation of the ShEx language based on the formal semantics presented in [4] and [5]. Our goal is to verify the correctness of the different lemmas, propositions and algorithms presented in those papers. We consider that this exercise can be useful as a first step to explore the suitability of automatic theorem provers to verify the correctness of the semantics of RDF validation languages. The proofs and definitions are based on the corresponding ones that appear in the papers.

The main contribution of the paper is a Lean 4 representation of the ShEx language as well as a verification of the correctness of the validation algorithms.

2 Related Work

Interactive theorem provers such as Lean 4 [14], Rocq (formerly Coq) [20] and Isabelle/HOL [15] allow definitions and proofs to be stated in a formal language and machine-checked, rather than only reviewed by hand. They have been used to formalize substantial bodies of mathematics and, increasingly, the semantics of programming languages and logics used in practice [7], as illustrated by verified systems such as the CompCert compiler [12,13], the seL4 microkernel [8], and by more recent industrial applications such as AWS's Cedar authorization language, formalized in Lean [6].

There have been several approaches to formally verify large parts of mathematics using interactive theorem provers like Lean, Rocq or Isabelle. The application of those techniques in the context of Knowledge Graphs and Semantic web is still not mature compared with other branches of mathematics. Some earlier work was developed using Isabelle/HOL to formalize OWL semantics in [21]. More recently, [19] verifies Datalog using Lean.

ShEx was initially introduced in [17] in an informal way based on an implementation. Later, [18] established the complexity of Shape Expressions without negation and presented a formal semantics for them based on regular bag expressions. An implementation based on regular expression derivatives was proposed in [11]. The formal semantics of the ShEx language including negation and recursion was presented in [4] and published as a W3C Community Group Report [16]. More recently, [5] presented an extension of ShEx with inheritance, which is currently developed by the IEEE P3330 Shape Expression Schemas Working Group. More recently, there was an interest in comparing ShEx with other validation approaches, such as SHACL [9] and PGSchema [3], and in formalizing the common foundations of these three languages: [1] formally compared ShEx, SHACL and PGSchema without considering recursion, and [2] compares ShEx and SHACL with recursion. None of those languages employed an automatic theorem prover to verify the correctness of their semantics and algorithms.

3 Representing RDF data in Lean

We define an RDF graph as a finite set of RDF triples: `abbrev Graph := Finset Triple` where a `Triple` is a structure:

```
structure Triple where
  subj : Subject
  pred : IRI
  obj  : RDFNode
```

such that an `RDFNode` is either an IRI, a literal or a blank node:

```
inductive RDFNode where
  | iri   : IRI → RDFNode
  | lit   : Literal → RDFNode
  | bnode : BNodeId → RDFNode

inductive Subject where
  | iri   : IRI → Subject
  | bnode : BNodeId → Subject
```

where IRI, Literal and BNodeId have a simplified definition as:

```
abbrev IRI := String
abbrev BNodeId := String
inductive Literal where
  | str : String → Literal
  | int : Int → Literal
```

A typing is a finite set of pairs of RDF nodes and shape labels:

```
abbrev Assoc  := RDFNode × Label
abbrev Typing := Finset Assoc
```

4 Representing ShEx abstract syntax and typings in Lean

The ShEx abstract syntax from [4] is represented in Lean as follows.

A ShEx schema is a finite set of shape labels and a function that maps each shape label to a shape expression:

```
structure Schema where
  labels   : Finset Label
  shapeDef : Label → ShExpr
```

A shape expression `ShExpr` is either a value description, a triple expression, or a boolean combination of shape expressions, and a triple expression `TExpr` is either a triple pattern, the empty triple expression, a choice or conjunction of triple expressions, or a repetition of a triple expression. Since a shape expression's neighbourhood description embeds a triple expression, and a triple pattern refers back to a shape label, the two are declared as a single mutually recursive block:

```
mutual
inductive ShExpr where
  | valueDescr : ValueDescr → ShExpr
  | neigDescr  : TExpr → ShExpr
  | and        : ShExpr → ShExpr → ShExpr
  | or         : ShExpr → ShExpr → ShExpr
  | not        : ShExpr → ShExpr

inductive TExpr where
  | tPattern : PropSet → Label → TExpr
  | empty    : TExpr
  | someOf   : TExpr → TExpr → TExpr
  | eachOf   : TExpr → TExpr → TExpr
  | rep      : TExpr → Nat → Option Nat → TExpr
end
```

A value description `ValueDescr` is a finite subset of RDF nodes, and a `PropSet` is a subset of properties (IRIs).

5 Representing semantic rules

The basic semantics of ShEx formalizes the `Satisfies` (⊢) and `Matches` (⊨) relations of Fig. 2 in [4]. both parameterized by a typing t. The two relations are mutually recursive, a shape expression's neighbourhood description invokes `Matches` on the node's outgoing triples, and a triple expression's shape reference

invokes `Satisfies` indirectly, through membership in `t` so they are declared together as a single mutual block, `Satisfies` and `Matches`, mirroring the paper's joint Fig. 2a/2b presentation.

Although `Satisfies` and `Matches` look like a classic mutually inductive derivability relation, they are encoded as ordinary Prop-valued recursive functions rather than a mutual inductive family. This is necessary because the premise in the rule involving negation is a negative occurrence of `Satisfies`, which Lean's positivity checker rejects for an inductive definition but accepts for a function.

`Satisfies` and `Matches` are defined case by case. Each equation corresponds to one (or a disjunction of two) rules.

```
mutual
def Satisfies (G : Graph) (t : Typing) (n : RDFNode) : ShExpr → Prop
  | .valueDescr V ⇒ InV n V ∨ BNodeInV n V
  | .neigDescr E ⇒ Matches G t (neigh G n) E
  | .and S1 S2 ⇒ Satisfies G t n S1 ∧ Satisfies G t n S2
  | .or S1 S2 ⇒ Satisfies G t n S1 ∨ Satisfies G t n S2
  | .not S ⇒ ¬ Satisfies G t n S

def Matches (G : Graph) (t : Typing) (N : Finset Triple) : TExpr → Prop
  | .tPattern P L ⇒
    ∃ subj pred obj, N = {(⟨subj, pred, obj⟩ : Triple)} ∧ pred ∈ P ∧ t.has obj L
  | .empty ⇒ N = ∅
  | .someOf E1 E2 ⇒ Matches G t N E1 ∨ Matches G t N E2
  | .eachOf E1 E2 ⇒
    ∃ N1 N2, DisjointUnion N N1 N2 ∧
             Matches G t N1 E1 ∧
             Matches G t N2 E2
  | .rep E min max ⇒
    ∃ Ns : List (Finset Triple),
      DisjointUnionList N Ns ∧
      min ≤ Ns.length ∧
      (∀ m', max = some m' → Ns.length ≤ m') ∧
      ∀ Ni ∈ Ns, Matches G t Ni E
end
```

`InV` is a proposition that passes if `n` is an RDF Node in `V`, `BNodeInV` passes if `n` is a blank node in `V`, and `neigh` is a function that returns the set of triples in `G` that have `n` as subject. `DisjointUnion` and `DisjointUnionList` are defined as follows:

```
def DisjointUnion (N N1 N2 : Finset Triple) : Prop :=
  Disjoint N1 N2 ∧ N = N1 ∪ N2

def DisjointUnionList (N : Finset Triple)
                      (Ns : List (Finset Triple)) : Prop :=
  Ns.Pairwise Disjoint ∧ N = Ns.foldr (· ∪ ·) ∅
```

6 Proving lemmas and propositions

6.1 Stratified schemas

Mixing unrestricted recursion with negation can make *correct typing* fail to find a unique semantics: the same schema can admit several incompatible correct

typings at once. A standard fix, in the spirit of stratified negation in logic programming, is to forbid negation from ever occurring on a recursive cycle between shape labels. This is captured by a dependency graph on shape labels with two kinds of edges: `DepPos S L1 L2` holds when the shape label L2 occurs as a shape reference in L1's definition without being under a NOT (a positive dependency), and `DepNeg S L1 L2` holds when L2 occurs under some NOT (a negative dependency).

```
abbrev DepPos (S : Schema) (L1 L2 : Label) : Prop := L2 ∈ (S.shapeDef L1).posDeps

abbrev DepNeg (S : Schema) (L1 L2 : Label) : Prop := L2 ∈ (S.shapeDef L1).negDeps
```

A schema with stratified negation admits a stratification `Strat`, represented as a structure that packages the number of strata `k`, the stratum `str` of each label, and proofs that this assignment respects the schema's dependency graph: negative dependencies must strictly decrease the stratum (`neg_strict`), while positive dependencies may never increase it (`pos_le`).

```
structure Strat (S : Schema) where
  k : ℕ
  str : Label → ℕ
  mem_range : ∀ L ∈ S.labels, 1 ≤ str L ∧ str L ≤ k
  neg_strict : ∀ L1 ∈ S.labels, ∀ L2, DepNeg S L1 L2 →
                 str L2 < str L1
  pos_le : ∀ L1 ∈ S.labels, ∀ L2, DepPos S L1 L2 →
                 str L2 ≤ str L1
```

`S` is a schema with stratified negation if it admits a stratification

```
def Schema.StratNeg (S : Schema) : Prop := Nonempty (Strat S)
```

6.2 Maximal Correct typing

Lemma 1 covers the base case where negation is entirely absent: for a negation-free schema `S` (`Schema.NegFree`, every label's definition uses no NOT) and any graph `G`, there is a correct typing that contains every other correct typing of `G` by `S`. The reason this case is easy is monotonicity: without negation, `Satisfies` only grows with the typing — a node satisfying a shape expression under some typing keeps satisfying it under any larger one — so simply taking the union of all correct typings (`Schema.maximalTyping`) is itself correct, and contains every other one by construction.

```
theorem lemma1 (S : Schema) (hnf : S.NegFree) (G : Graph) :
    ∃ typingG : Typing, typingG ⊆ S.univTyping G ∧ S.CorrTyping G typingG ∧
      ∀ t', t' ⊆ S.univTyping G → S.CorrTyping G t' → t' ⊆ typingG
```

It is possible to construct a stratified-maximal correct typing `MTyping` one stratum at a time.

```
def MTyping (G : Graph) (S : Schema) (str : Strat S) : Typing :=
  typ S str.str G str.k

def typ (S : Schema) (str : Label → ℕ) (G : Graph) : ℕ → Typing
  | 0 ⇒ ∅
  | i + 1 ⇒ maxAgree (S.stratRestrict str (i + 1)) G
                     (S.stratum str i) (typ S str G i)
```

where `maxAgree S' G Lo fixed` is the union of all correct typings of `G` by `S'` that agree with a given typing fixed on the labels in `Lo` — the per-stratum construction that `typ`'s inductive step performs, generalizing Lemma 1's maximal correct typing (the special case `Lo = ∅`, no agreement constraint).

For any schema `S`, any stratification `str` of `S` and any graph `G`, the `MTyping G S str` is a correct typing of `G` by `S`.

```
theorem proposition1 (S : Schema) (str : Strat S)
                     (hClosed : S.Closed) (G : Graph) :
    S.CorrTyping G (MTyping G S str)
```

The Lean proof strengthens the statement before inducting: rather than reasoning about MTyping directly, it shows by induction on the stratum index i that `typ S str G i` satisfies two invariants at once — it stays within S_i's universe of typings, and it is already a correct typing of S_i. The base case ($i = 0$) is vacuous, since the initial stratum L_0 is empty. The inductive step reduces to a single correctness fact about `maxAgree` (the union-of-typings construction underlying `typ`), applied with the previous stratum as the "agreement set" — well-formedness of the schema together with the stratification's `neg_strict` condition is exactly what guarantees that S_{i+1}'s negative dependencies stay confined there. Proposition 1 is then just this invariant read off at the last stratum, $i = k$.

`hClosed` is a well-formedness hypothesis with no counterpart in the paper's own statement of Proposition 1: it requires every shape label referenced inside `S`'s definitions to itself be declared. In the paper this holds automatically, since schemas are only ever defined on their own label set; in Lean, `shapeDef` must be a total function, so this closure condition has to be assumed explicitly.

Lemma 2 states that for any two stratifications `str1`, `str2` of a schema `S` the maximal correct typing does not depend on which stratification was chosen:

```
theorem lemma2 (S : Schema) (hClosed : S.Closed)
               (str1 str2 : Strat S)
               (G : Graph) : MTyping G S str1 = MTyping G S str2
```

Here the paper and the Lean development diverge more than for Proposition 1: the paper's own proof is stated only as an idea, asserting without further justification that a canonical *most refined* stratification exists which every other stratification agrees with. The Lean proof has to supply this argument in full. It introduces a canonical process that starts from the empty typing and repeatedly reveals the least-stratum batch of labels still remaining; an *Independence* argument then shows, by strong induction on the number of labels left, that the process's final result does not depend on the order in which batches are revealed — the technical heart being a *Swap* lemma letting two mutually-independent batches be revealed in either order. Running this process along the checkpoints of an arbitrary stratification shows it agrees with `MTyping` for that stratification; applying this to `str1` and `str2` separately, both equal the same canonical value, giving Lemma 2.

7 Validation algorithms

The previous definitions characterize the correct semantics of a schema declaratively as the union of correct typings that a stratification happens to pick out, but say nothing about how to compute it. The paper gives two validation algorithms and states their correctness with respect to that declarative semantics: Algorithm 1 (*refine*), which computes the whole typing `MTyping G S str` at once, and Algorithm 2 (*prove*), which decides membership of a single node-label pair without ever constructing the whole typing. Both algorithms, and both correctness statements, have been fully formalized and machine-checked in Lean.

7.1 Algorithm 1: refine

Refine computes one stratum at a time: for stratum i, it starts from the largest plausible typing (everything already fixed for stratum $i-1$, plus every possible guess for the new labels of stratum i), then repeatedly strips out any node-type association that currently fails `Satisfies`, until nothing more can be removed — a greatest-fixed-point computation, one per stratum, chained together.

```
def Algorithm1 (G : Graph) (S : Schema) (str : Strat S) : Typing :=
  refineAlg S str.str G str.k

theorem algorithm1_correct (G : Graph) (S : Schema) (str : Strat S)
    (hClosed : S.Closed) : Algorithm1 G S str = MTyping G S str
```

The Lean proof mirrors this stratum-by-stratum structure directly. A single refinement pass (`refineStep`) is iterated to a fixed point (`refine1`), and a core lemma (`refine1\_eq\_maxAgree`) shows that this iterative fixed point computes exactly the same typing as `maxAgree` — the declarative, union-of-correct-typings construction that `typ`'s own inductive step uses. Chaining this per-stratum equivalence across all k strata by induction then gives `algorithm1\_correct`: two very different-looking constructions, iterative refinement versus a declarative union, turn out to compute the same object at every stratum.

7.2 Algorithm 2: prove

Prove decides whether (n, L) belongs to the typing without ever constructing it, by recursively visiting only the portion of the dependency graph reachable from (n, L), tracking a stack of node-label pairs currently being resolved (to detect and trust positive cycles) and a set `Dep` of pairs it has confirmed along the way.

```
def prove (S : Schema) (G : Graph) (n : RDFNode) (L : Label) : Bool :=
  proveAux S G (G.nodes ×ˢ S.labels) n L ∅

theorem prove_correct (S : Schema) (str : Strat S) (hClosed : S.Closed)
    (G : Graph) (n : RDFNode) (L : Label) (hLlabels : L ∈ S.labels) :
    prove S G n L = true ↔ (n, L) ∈ MTyping G S str
```

This proof is structurally the more delicate of the two. `proveAux` is a *coinductive-style* algorithm — when it re-encounters a pair already on the call

stack, it simply trusts it, which is exactly what makes positive recursive cycles terminate — so relating it to `MTyping`'s greatest-fixed-point characterization needs a genuine witness typing, not just an unfolding argument. The proof first strengthens the semantic relations `Satisfies`/`Matches` to a *neighborlocal* form, showing that whether (n, L) holds depends on the ambient typing only through its restriction to (n, L)'s own dependency set — the fact that lets `prove` get away with tracking `Dep`/`Hyp` instead of a whole typing. Soundness and completeness are then proved together, by strong induction on the stratum, as a single strengthened statement (`MainProp`) relating `proveAux`'s call-stack bookkeeping to membership in `typ` at that stratum:

```
def MainProp (S : Schema) (str : Strat S) (G : Graph) (U : Finset Assoc)
    (i : ℕ) : Prop :=
  ∀ Hyp : Finset Assoc,
    (∀ p ∈ Hyp, p.2 ∈ S.stratum str.str i → p ∈ typ S str.str G i) →
    ∀ n L, str.str L = i → L ∈ S.labels →
      (proveAux S G U n L Hyp = true ↔ (n, L) ∈ typ S str.str G i)
```

The full argument runs several hundred lines in `Algorithm2.lean`; we refer to it, and to §4.2 of the paper for the algorithm's original presentation, rather than reproduce it here.

8 Adding inheritance

The latest version of ShEx added the notion of shape *inheritance* and was presented in [5]: a shape can `extend` one or more other (abstract) labels, and validation against an abstract label succeeds if the node matches *some* concrete descendant of it. Rather than retrofit this onto the 2017 formalization, we ported it as a genuine second formalism, `ShexLean.Inheritance`, with its own `ShapeExpr`/`Schema`, reusing some common parts like (`Graph`, `Typing` and `Label`). The syntax itself grows in two ways beyond what 2017 needed: node constraints become an arbitrary boolean predicate rather than a finite `ValueDescr`, and `closed`/`extra` shape modifiers and **`extends`** become first-class:

```
structure Shape where
  closed : Bool
  extra  : Option (Finset IRI)
  triples : TripleExpr

inductive ShapeExpr where
  | ofShape (h : Shape) : ShapeExpr
  | extends (xs : Finset Label) (h : Shape) : ShapeExpr
  | constraint (c : NodeConstraint) : ShapeExpr
  | or (s1 s2 : ShapeExpr) : ShapeExpr
  | and (s1 s2 : ShapeExpr) : ShapeExpr
  | not (s : ShapeExpr) : ShapeExpr
```

The 'extends' case is what breaks the 2017 proof pattern at its foundation. In the 2017 semantics, a shape reference is resolved purely by consulting the typing, so `Satisfies`/`Matches` are well-founded by ordinary structural recursion on the finite syntax tree. An 'extends' clause instead recurses into *other labels' own*

definitions, which implies a schema lookup, not a subterm of the current expression, so a plain structurally-recursive Lean definition is not just inconvenient here, it would be outright *false* if the schema's extension hierarchy happened to be cyclic. The paper handles this with its own Lemma 1 (termination, proved separately from the semantics), so the Lean development follows the same shape: `Satisfies` is first defined as a total, `Fuel`-indexed under-approximation:

```
def SatisfiesFuel : (Fuel : ℕ) → Schema → Graph → Typing → RDFNode →
    ShapeExpr → Prop
  | 0, _, _, _, _, _ ⇒ False
  ...
```

and a separate file, `Termination.lean`, proves that for a well-defined schema some large-enough, canonical fuel value makes the answer stop depending on `Fuel` at all, recovering the paper's real, fuel-free `Satisfies`. Turning the paper's one-sentence termination argument ("the tree of recursive calls is finite due to the acyclic nature of the extension hierarchy") into a Lean proof needs an actual rank function on labels witnessing that acyclicity, combined into a single measure that decreases on both kinds of recursive step: ordinary AST descent and 'extends'-hops.

Definition 4 (correct typing) also gains a genuinely new case beyond a direct port of the 2017 statement: an abstract label is satisfied not by its own definition, but by the *existence* of some concrete descendant that is:

```
def Schema.CorrectTyping (S : Schema) (G : Graph) (τ : Typing) : Prop :=
  ∀ n z, τ.has n z →
    (z ∉ S.A → Satisfies S G τ n (S.shapeDef z)) ∧
    (z ∈ S.A → ∃ x ∈ S.desc z \ S.A, Satisfies S G τ n (S.shapeDef x))
```

From there, Proposition 1, Lemma 2, and Algorithm 1 are structurally direct ports of their 2017 counterparts — same per-stratum greatest-fixed-point idea for the algorithm, same Fusion/Swap/Independence argument for Lemma 2 (whose own paper proof, again, is only sketched as "Proof. (Idea)...") — but none of them transfer *mechanically*. The 2017 monotonicity and schema-agreement lemmas these all lean on go through by plain structural recursion, because 2017's 'Satisfies' never dereferences the schema. Inheritance's 'Satisfies' does, via ancestor lookups, so every one of those lemmas first had to be re-proved against the schema's dependency structure and then generalized from a single stratification-tied restriction ('stratRestrict') to an arbitrary dependency-closed label set ('Schema.restrictLabels'), before Lemma 2's schedule-agnostic argument and Algorithm 1's per-stratum schema-swapping could reuse it. Algorithm 2 (*prove*) has not yet been ported to this formalism.

In order to formalize the definitions we needed to tackle some potential places in which the paper didn't clearly specify all the conditions. For example, Table 3's `extra` rule is *not monotone* in the typing without a further condition: growing the typing can move a triple out of the "extra-covered" bucket and into the main triple-expression match, which can break a bare (non-starred) triple constraint that needed an exact singleton. We checked a concrete instance of this directly in Lean: a shape with `extra := some \{p\}` and `triples := p @z`, satisfied

under a typing τ_1, provably *stops* being satisfied under a strictly larger $\tau_2 \supseteq \tau_1$. The appendix proof of Proposition 1 in the paper states the fix for this: an invariant it calls "(*)": a shape's triple-constraint properties never overlap its own `extra` set. That definition was not defined as a standalone definition in the main text: we named it explicitly as `Schema.WellFormedExtra` and added it as a hypothesis everywhere Proposition 1 and its descendants need it.

The Lean development is available at `https://anonymous.4open.science/r/shex_lean-B1A5/` and it contains:

- the 2017 semantics and its proofs,
- the inheritance extension and its proofs
- all the examples from both papers associated with their corresponding numbers in the paper
- an executable main with two command-line options: `list` to show all the propositions and examples that can be checked, and `run` to check a specific one by its name.

9 Conclusions and future work

Using Lean, it is possible to automatically verify the different lemmas that relate the semantics of the ShEx languages. The resulting repository can be used as a first approach to explore more design decisions that extend the ShEx language. Future work includes a more extensive exploration of the design space for RDF validation languages including extending ShEx for RDF 1.2 features or other language features.

Declaration of use of Generative AI

The authors declare that the content of this paper and the accompanying Lean code was developed with the assistance of Claude Code (Sonnet 5), ChatGPT and GitHub Copilot. The authors have reviewed and edited the content as needed and take full responsibility for the content of this publication.

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10 Appendix: Abstract syntax of ShEx with inheritance

Definition 1 of [5] fixes two disjoint finite sets of shape labels, Y (*simple* labels) and X (*extendable* labels). A shape expression over Y and X is either a reference `@z`, or a plain shape expression derivable from the non-terminal u in the following grammar:

```
u ::= h | t | c | u or u | u and u | not u
h ::= [closed] [extra P] <e>
t ::= extends X h
c ::= a boolean function over IRI ∪ Blank ∪ Literal
e ::= ε | p @z | e|e | e;e | e*
```

where u is a plain shape expression, h a shape, t a shape with extends, c a node constraint, and e a triple expression; $z \in Y \cup X$, $X \subseteq X$, $p \in IRI$, $P \subseteq IRI$, the symbol `ε` is a constant, and `@`, `|`, `*` and `;` are operators.

This grammar corresponds directly to the Lean types introduced in Section 8: `TripleExpr` encodes e (constructors `empty`, `tc`, `choice`, `seq`, `star`), and `ShapeExpr` encodes u (constructors `ofShape`, `extends`, `constraint`, `or`, `and`, `not`). The `@z` reference is not modeled as its own constructor: it only ever occurs nested inside a triple constraint `p @z`, so it is represented directly by the `tc` constructor of `TripleExpr` rather than by a recursive case of `ShapeExpr`.